Syllabus

2025 Fall

Course	Software Engineering	Professor	Abhijit Tarawade
Course No	CSE4070	Class No	
Schedule	SAT9	Grading Eval.	Relative Evaluation
Other Information			

Profile	**Prof.(Dr.)Abhijit Tarawade** **office A402-1** **email a.tarawade@inha.uz**
Course Objectives	This course allows students to understand the software process and product, basic techniques for software development, including analysis, design, construction, testing, and maintenance, and software project management and quality assurance. This course suggests students to provide a course project. The project allows students to take a part in a development of a practical software application. Especially, several logistics software applications such as SCM, ERP, and so on, are introduced and logistics examples are modeled
Course Description	
Textbooks	Title of Publications:Software Engineering : A P R A C T I T I O N E R'S A P P R O A C H Author: Roger Pressman Publication Company: McGraw Hill Publication Year: 2009
Other Texts and References	
Class Structure	
Notes	
ABEEK	

Grading

Mid-term	Final exam	Attendance	Assignments	Quiz	Discussion	ETC	Total
30 %	30 %	10 %	0 %	10 %	20 %	0 %	100 %

S y l l a b u s

Week	Content	Class	Notes
1	Theme	The Product and Process	Readings: Chapters 1 - 4.
	Class Details	An overview of software engineering and process Discussion of project requirements and topics	
	Tests		
2	Theme	Project Planning and Organization	Readings: Chapters 5. 24, 25, 26
	Class Details	Project planning issues Software estimation techniques Risk analysis	
	Tests	Project Abstract due	
3	Theme	Software Engineering Practice	Readings: Chapters 5, 6
	Class Details		
	Tests		
4	Theme	Requirements Engineering	*Readings:* Chapters 7
	Class Details		

	Tests		

5	**Theme**	Analysis Methods	Readings: Chapter 8
	Class Details	UML notation	
	Tests		
6	**Theme**	Elements of software design	Readings: Chapter 9, 14
	Class Details	Fundamental concepts The design model Design patterns	
	Tests		
7	**Theme**	**Midterm exam**	
	Class Details		
	Tests	Requirements specification due	
8	**Theme**	Design Methods - I	Readings: Chapters 10, 11
	Class Details	Architecture and component design	
	Tests		
9	**Theme**	Design Methods - II	Readings: Chapters 12, 13
	Class Details	User experience design and mobility design	
	Tests		
10	**Theme**	Testing and Reviews	Readings: Chapters 16, 19
	Class Details	Reviews and inspections White-Box and Black-box testing methods	
	Tests		
11	**Theme**	Testing Methods	Readings: Chapter 20, 21
	Class Details	Integration testing Specialized testing (e.g. mobility and security)	
	Tests	Test Specification due	
12	**Theme**	Umbrella Activities - 1	Readings: Chapters 15, 17
	Class Details	Quality Concept Software Quality Assurance	
	Tests		
13	**Theme**	Umbrella Activities - 2	Readings: Chapters 18
	Class Details	Software Security Engineering	
	Tests		
14	**Theme**	Umbrella Activities - 3	Readings: Chapters 22
	Class Details	Software Configuration Management	
	Tests		
15	**Theme**	Final exam	
	Class Details		
	Tests		
16	**Theme**	Make Up Class	
	Class Details		
	Tests		